RateMyResume

Enterprise On-Premises Architecture

Fully on-premises deployment with GPU-based AI processing and a scale-out path

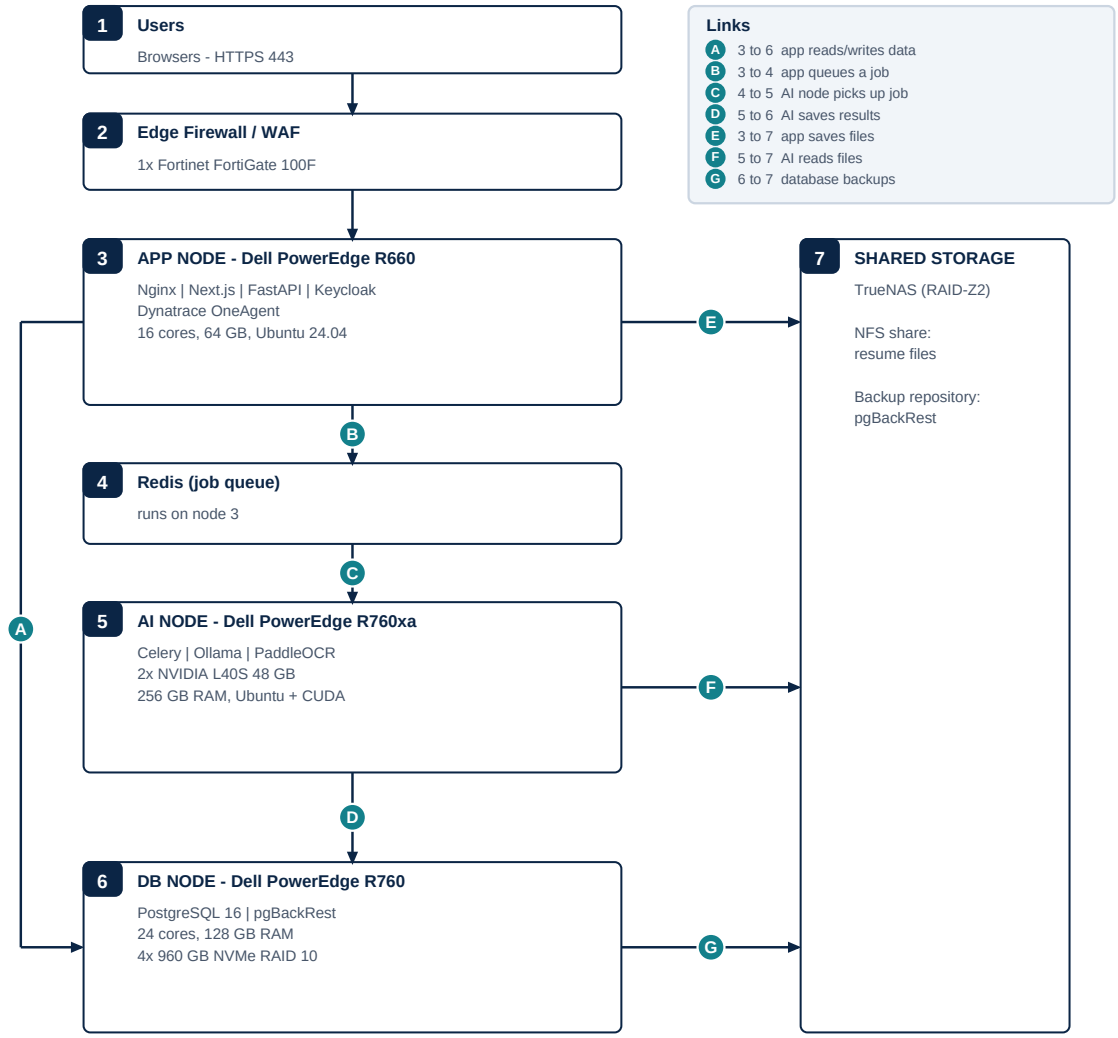
Version 2.3 - simplified design

1. Overview and Design Basis

This document describes the on-premises architecture for RateMyResume. The initial deployment uses **one node per tier**, which keeps cost and complexity low. Section 4 explains how to add nodes as traffic grows, which also adds high availability and automatic failover.

Requirement	Design decision
Load: about 5,000 resumes per day	Roughly 10 per minute on average. One app node, one AI node and one database node are the starting point. Benchmark with your chosen model before final sizing.
Larger LLMs	Two NVIDIA L40S (48 GB) in the AI node (5). Quantized models up to roughly 70B parameters fit.
Fully on-premises	No cloud dependencies for the application. Models run locally on Ollama; resume data stays on site.
Monitoring	Dynatrace OneAgent on the app node 3 (see Section 5).

2. Architecture Diagram



Scale-out: add more of nodes 3, 5 and 6 as traffic grows (Section 4).
Network: 1x 25GbE switch, VLANs app / ai / data / storage / management.

3. Hardware and Software

Entity	Hardware	Software
1. Users	Desktop, laptop or mobile	Any modern browser
2. Edge firewall / WAF	1x Fortinet FortiGate 100F (or reuse your existing firewall)	FortiOS: IPS, WAF, TLS inspection
3. App node (hosts 4. Redis)	1x Dell PowerEdge R660 (1U). 16 cores, 64 GB RAM, 2x 480 GB SSD RAID 1	Ubuntu Server 24.04 LTS, Nginx, Next.js, FastAPI, Keycloak, Redis, Dynatrace OneAgent
5. AI node	1x Dell PowerEdge R760xa (2U). 256 GB RAM, 2x 1 TB NVMe, 2x NVIDIA L40S 48 GB	Ubuntu 24.04 + NVIDIA CUDA, Celery, Ollama, PaddleOCR
6. DB node	1x Dell PowerEdge R760 (2U). 24 cores, 128 GB RAM, 4x 960 GB NVMe RAID 10	Ubuntu 24.04, PostgreSQL 16, pgBackRest
7. Shared storage (NAS)	TrueNAS (or equivalent NAS), RAID-Z2	NFS share for resume files; backup repository
Network	1x 25GbE switch (add a second for redundancy)	VLANs: app, ai, data, storage, management

Hardware models and sizes are recommendations. Confirm availability, GPU compatibility and pricing with your vendor.

How the numbered entities connect

Link	From to	What happens
A	3 to 6	FastAPI reads and writes users, resumes and results directly in PostgreSQL (port 5432).
B	3 to 4	FastAPI puts an extraction or analysis job in the Redis queue (6379) and replies to the user at once.
C	4 to 5	Celery workers on the AI node pick jobs from the queue. The app never calls the AI node directly.
D	5 to 6	The AI node saves OCR text and analysis results straight into PostgreSQL.
E	3 to 7	FastAPI saves uploaded resume files on the NAS share (NFS 2049).
F	5 to 7	The AI node reads those files for OCR and analysis.
G	6 to 7	pgBackRest writes nightly backups and the WAL archive to the NAS.

Upload flow in one paragraph

1 uploads over HTTPS through 2 to 3. Node 3 saves the file to 7 (E), records the resume in 6 (A) and queues a job in 4 (B), then replies immediately. Node 5 picks up the job (C), reads the file (F) and writes the results to 6 (D). The page then shows the result, read from 6 by 3 (A).

4. Scaling Out and High Availability

With one node per tier, a server failure stops that tier until it is repaired. Add the following as traffic or availability needs grow. The application code does not change, because all nodes share the same queue and database.

Add	What it gives	Also needed
Second app node (3)	Web/API capacity and failover	HAProxy + keepalived (shared virtual IP), Redis replica + Sentinel, Keycloak cluster
Second AI node (5, same spec)	More resumes per hour; AI redundancy	Nothing: it consumes the same Redis queue
DB replica + witness (6)	Automatic database failover	Patroni, etcd (3 members, including a small witness server), HAProxy for routing
Second firewall (2), switch, NAS controller (7)	Removes remaining single points	HA firewall pair, MLAG switch pair, dual-controller NAS

If this fails	Initial deployment	After scale-out
3. App node	Web/API unavailable	Other app node serves traffic
5. AI node	Uploads work; analysis waits in the queue	Other AI node continues
6. DB node	Service down; restore from backup	Replica promoted automatically
7. NAS	File access stops	Dual controller covers controller failure

5. Monitoring with Dynatrace

Yes, you can install Dynatrace on the app node (3). The Dynatrace OneAgent runs on the same host as the application and needs no extra server. Points to note:

- The agent only collects data. The Dynatrace backend that stores and analyses it is separate: either Dynatrace SaaS (cloud, which conflicts with a fully on-premises requirement) or a self-hosted Dynatrace Managed cluster, which needs its own servers. Confirm current options with Dynatrace.
- The agent uses some CPU and memory on the app server, so leave headroom.
- Dynatrace is licensed. If the cost is a concern, Prometheus and Grafana are free alternatives that run on a small server.
- You can install the same agent on the AI and DB nodes later for full visibility.

6. Security

- Separate VLANs with deny-by-default firewall rules between tiers.
- TLS on external and internal links; encryption at rest for database and storage volumes.
- Keycloak issues short-lived JWTs; FastAPI enforces per-user ownership of resumes.
- Resumes contain personal data: keep inference local, avoid logging resume content, apply retention policies.
- Secrets in a vault, restricted admin access, regular backup restore tests.

7. Open Items

- **Ollama concurrency:** it handles limited parallel requests per model. Benchmark and switch to vLLM if throughput is short.
- **Model size:** each GPU holds 48 GB; models beyond about 70B quantized need more GPUs.
- **Vendor checks:** L40S support in the R760xa, and firewall throughput with TLS inspection.
- **Load test:** validate at 5,000 resumes per day with the chosen model before go-live.